REPUBLIC POLYTECHNIC

C339 Software Testing and Analytics
Continuous Assessment 1 (CA1)

Project 05: Grade Checking System Requirements Specification

1. About

A grade service requires a simple software system to calculate the average of two marks and determine whether a recorded score is a pass.

The system consists of two independent functions:

- **Calculate Average Mark:** Calculate the arithmetic mean of two equally weighted marks.
- **Determine Pass Status:** Determine whether a recorded score meets the pass mark.

Both functions must operate according to the requirements described below.

2 General Requirements

The following requirements apply to both functions:

1. All marks and scores are whole numbers from 0 to 100 inclusive. The calculated average may include a fractional part.
2. Numeric inputs must be Python integers (int). Boolean values are not valid integers. A parameter explicitly declared as bool accepts only True or False.
3. If an argument has an invalid data type, the function must raise a **TypeError**.
4. If an argument has the correct data type but falls outside the permitted range, the function must raise a **ValueError**.
5. The functions operate independently and do not require a database, network connection, or shared application state.

3 Functional Requirements

3.1 Calculate Average Mark

Source File: grade_average.py

Function: calculate_average(mark_one, mark_two)

Description

Calculate the arithmetic mean of two equally weighted marks.

Parameter	Data Type	Constraints
mark_one	int	Must be an integer from 0 to 100 inclusive
mark_two	int	Must be an integer from 0 to 100 inclusive

Requirements

1. Add mark_one and mark_two, then divide the sum by 2 using ordinary division.
2. Marks of 0 and 100 are allowed. Keep any fractional part; do not round down.

Output: Return the numerical average. For example, marks of 50 and 51 have an average of 50.5.

3.2 Determine Pass Status

Source File: grade_pass.py

Function: has_passed(score)

Description

Determine whether a recorded score meets the pass mark.

Parameter	Data Type	Constraints
score	int	Must be an integer from 0 to 100 inclusive

Requirements

1. A score of 50 or more is a pass.
2. A score below 50 is not a pass. A missing score such as None is an invalid type.

Output: Return True for a pass and False otherwise. Return a Python bool.